Souvik Sarkar

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EDUCATION

National Institute of Technology Durgapur 2023 – 2027
B.Tech. in Computer Science and Engineering – **CGPA: 8.66** Durgapur, West Bengal

EXPERIENCE

Pinggy Technology May 2026 – Present
Software Development Engineer Intern Remote

- Developed 3 autonomous Python agents using **Playwright (Patchright), FastAPI and Chrome DevTools Protocol (CDP)** to simultaneously syndicate engineering blogs across multiple platforms, eliminating manual publishing overhead and Integrated **Gemini and OpenRouter APIs** to rephrase blog posts.
- Designed a robust error-handling mechanism featuring human handoff points to bypass dynamic authentication challenges (CAPTCHAs/OTPs) without crashing execution loops.
- Architected and built a multi-tenant Social Media outreach platform featuring a **React (Redux Toolkit/RTK Query)** frontend, a **Java 21 (Spring Boot 3)** backend orchestrating campaign queues concurrently via **PostgreSQL** row-locking and a stateless **Python 3.12 (FastAPI)** worker that executes browser automation workflows with anti-detection pacing.

Jadavpur University Dec 2025 – Jan 2026
Research Intern — Guide- Prof. Debotosh Bhattacharjee Remote

- Developed a deep learning model for medical image anomaly detection, implementing a **Teacher- Student architecture with PyTorch** for medical imaging (e.g., Head CT scans), utilizing a pre-trained **Wide ResNet-50-2** as the teacher and a custom-designed DeResNet decoder as the student.
- Designed and integrated multi-scale skip connections and a custom **Multi-scale Anomaly Consistency (MAC) loss** (combining Cosine Similarity and MSE) to align intermediate representations and enforce reconstruction consistency across multiple spatial scales achieving **AUROC of 0.86** and **F1-score of 0.91**.

PROJECTS

Audio Classifier Visualizer **Live**
PyTorch, Torchaudio, TensorBoard, Modal, React, Next.js, TypeScript, Tailwind CSS

- Architected a **ResNet-style audio classification CNN** using the ESC-50 dataset, achieving **84.75% accuracy**.
- Processed raw waveforms into log-amplitude **Mel-spectrograms** for robust feature extraction.
- Built an interactive dashboard for audio uploads, waveform visualization, and real-time inference.

MediBot AI **Live**
Python, LangChain, HuggingFace, Pinecone, Flask, Docker, Sarvam API

- Engineered a **Medical Retrieval-Augmented Generation (RAG) system** over custom medical books.
- Designed a hybrid dual-engine architecture with **Sarvam AI** for multilingual healthcare assistance.

StyleHatch **Live**
React, MongoDB, Express, Node.js, Tailwind CSS, Redis, Vite.js, Stripe

- Developed a **full-stack e-commerce platform** with authentication, product management and secure checkout workflows.
- Integrated **Stripe payment gateway** and **Redis-based optimizations** for improved performance.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, SQL

Technologies: React, Node.js, Express.js, MongoDB, MySQL, PyTorch, NumPy, Pandas, Matplotlib, Scikit-learn

Tools: Git, GitHub, Linux, VS Code, Postman, Eclipse, MS Excel

Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, Artificial Intelligence and Machine Learning

CODING PROFILES

- Solved **1000+ problems** across coding platforms.
- **LeetCode: Knight** (souvik31, highest rating: 1853), **CodeChef: 3-Star** (Souvik_31, highest rating: 1743), **Codeforces: Specialist** (souvik_31, highest rating: 1406).

ACHIEVEMENTS

- Secured **99.02 percentile** in JEE Main 2023.
- Secured **AIR 1486** in WBJEE 2023 among 1 lakh+ candidates.
- OCI 2025 Certified Foundations Associate. Certificate